

Hybrid Spatial-Temporal Modeling for Violence Detection: Combining CNNs and Video Transformers

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Abstract

Automated violence detection in video data is a challenging problem in computer vision, as violent behavior often emerges from subtle temporal interactions and can be visually similar to non-violent activities such as sports or physical play. Real-world video footage further complicates this task due to variations in camera motion, illumination, video quality, and recording conditions.

Existing approaches to violence detection have relied on either hand-crafted motion features or fully spatiotemporal deep learning models such as 3D convolutional neural networks and video transformers. While effective in controlled settings, these methods are often computationally expensive, require large labeled datasets, and are susceptible to short-cut learning caused by dataset-specific visual biases, limiting their robustness and scalability.

In this work, we propose a hybrid spatial-temporal architecture that decouples visual feature extraction from temporal reasoning. Frame-level semantic representations are extracted using a frozen ResNet50 backbone pretrained on ImageNet, reducing training complexity and computational cost. Temporal dynamics are modeled using a lightweight Transformer Encoder operating on compact frame embeddings augmented with sinusoidal positional encoding, enabling the capture of long-range temporal dependencies while efficiently handling variable-length video sequences. A masked pooling strategy aggregates temporal information into a fixed-length video representation for classification.

We evaluate our approach on the Real Life Violence Situations Dataset and achieve strong in-distribution performance, with an F1-score of 0.963 and a balanced accuracy of 0.962 on a held-out test set. Cross-dataset evaluation highlights sensitivity to domain shift, indicating that data diversity remains a key factor for robust violence detection. Overall, the proposed framework demonstrates an effective balance between performance, efficiency, and architectural flexibility.

Introduction

Automated violence detection in video streams has emerged as an important research problem at the intersection of computer vision and public safety. The task aims to identify aggressive or violent human behavior in unconstrained video footage, such as surveillance recordings or user-generated content, in order to support early intervention and situational awareness. Despite significant progress in video understanding, reliably detecting violence remains challenging due to

the subtle and context-dependent nature of human interactions, variations in camera motion, illumination, and recording quality, as well as the high visual similarity between violent actions and non-violent activities such as sports or physical play.

The importance of this problem is underscored by the growing availability of video data in public and semi-public spaces, where manual monitoring is infeasible at scale. Automated violence detection systems have the potential to support human operators by filtering large volumes of video streams and highlighting potentially critical events. At the same time, the societal impact of false alarms and missed detections is substantial, making robustness and balanced decision-making essential for real-world applicability.

Early approaches to violence detection relied on hand-crafted motion descriptors and optical flow statistics, which provided limited robustness under real-world conditions. With the advent of deep learning, spatiotemporal convolutional architectures such as 3D-CNNs enabled joint modeling of spatial appearance and temporal dynamics. While effective, these models are computationally expensive and often require large amounts of labeled video data, limiting their scalability and deployment in resource-constrained settings. More recently, attention-based models and video transformers have demonstrated strong performance by explicitly modeling long-range temporal dependencies, but full spatiotemporal transformers remain costly and sensitive to dataset biases.

In this work, we build on recent hybrid approaches that decouple spatial and temporal modeling. Specifically, we propose a modular CNN–Transformer architecture that combines a frozen ResNet50 backbone for frame-level feature extraction with a lightweight Transformer Encoder for temporal reasoning. By leveraging strong visual representations learned from large-scale image datasets and applying self-attention over compact frame embeddings, our method achieves an effective balance between expressive power and computational efficiency. The proposed design enables flexible handling of variable-length video sequences while maintaining robustness to domain shifts and ambiguous non-violent interactions, making it well-suited for real-world violence detection scenarios.

Related Work

The field of automated violence detection has evolved from classical computer vision techniques to sophisticated deep learning architectures capable of modeling complex spatial and temporal dynamics.

Application’s Perspective: Violence Detection

Early approaches primarily relied on hand-crafted features. A seminal work is the Violent Flows (ViF) descriptor proposed by Hassner, Itcher, and Kliper-Gross (2012), which utilized local optical flow changes to detect aggression. While efficient, these methods struggle with illumination changes and camera motion.

With the deep learning shift, researchers adopted 3D Convolutional Neural Networks (3D-CNNs) like C3D (Tran et al. 2015) and I3D (Carreira and Zisserman 2017). These models extend 2D convolutions into the temporal dimension to extract spatiotemporal features simultaneously. However, 3D-CNNs entail high computational costs and parameter counts, posing challenges for deployment in resource-constrained environments.

Methodology’s Perspective: CNN-Transformer Hybrids

To mitigate the limitations of 3D-CNNs, recent literature explores hybrid architectures that decouple spatial feature extraction from temporal modeling. The ResNet architecture (He et al. 2016) has become the standard backbone for spatial feature extraction, utilizing residual connections to enable the training of deep networks without vanishing gradient issues.

For temporal reasoning, the Transformer architecture, originally designed for NLP (Vaswani et al. 2017), has been adapted for video analysis. Unlike Recurrent Neural Networks (RNNs), Transformers utilize a self-attention mechanism to weigh frame importance globally. Works like TimeSformer (Bertasius, Wang, and Torresani 2021) and ViViT (Arnab et al. 2021) demonstrate state-of-the-art performance using spatiotemporal tokens. Alternatively, hybrid approaches combining pre-trained convolutional backbones with transformer encoders (Girdhar et al. 2019) offer a computationally efficient solution. Our work follows this paradigm by leveraging a ResNet50 backbone and a Transformer Encoder for classification.

Methodology

Overview of the Proposed Pipeline

We propose a modular, computationally efficient video-based violence detection framework that decomposes the spatiotemporal understanding task into two complementary stages: (i) frame-level semantic feature extraction using a frozen convolutional backbone, and (ii) long-range temporal reasoning using a transformer-based encoder. This design enables the model to leverage strong visual representations learned from large-scale image datasets while maintaining flexibility and efficiency in modeling variable-length video sequences.

Given an input video $V = \{I_1, I_2, \dots, I_T\}$ consisting of T RGB frames, the pipeline maps each frame to a high-dimensional embedding, models temporal dependencies across the resulting sequence, aggregates the temporal representations into a fixed-length video-level descriptor, and produces a binary violence prediction via a nonlinear classification head. The overall architecture is illustrated in Figure 1.

Video Representation and Frame-Level Feature Extraction

Each video is represented as a sequence of frames of varying length sampled at fixed temporal intervals. Rather than operating directly on raw pixel sequences using spatiotemporal convolutions, we decouple spatial and temporal modeling by first embedding each frame independently into a semantic feature space. This design reduces computational complexity and allows the reuse of robust visual representations learned from large-scale image classification tasks.

Let $I_t \in R^{H \times W \times 3}$ denote the t -th frame of a video. A pretrained ResNet50 convolutional backbone ϕ_{CNN} is used to extract a high-level feature vector:

$$\mathbf{f}_t = \phi_{\text{CNN}}(I_t), \quad \mathbf{f}_t \in R^{2048} \quad (1)$$

The final classification layer of the backbone is removed, and all convolutional parameters are kept frozen during training. This strategy stabilizes optimization on relatively small video datasets and constrains the learning process to the temporal modeling and classification components, which are more directly aligned with the violence detection task.

Feature Projection and Dimensionality Reduction

To reduce the computational burden of the subsequent temporal encoder and to mitigate overfitting, the high-dimensional frame embeddings are passed through a learned linear projection layer:

$$\mathbf{z}_t = W_p \mathbf{f}_t, \quad \mathbf{z}_t \in R^{256} \quad (2)$$

where $W_p \in R^{256 \times 2048}$ is a trainable projection matrix and the output dimensionality is fixed to 256. This transformation serves as an information bottleneck that encourages the model to retain task-relevant semantics while discarding redundant or noisy visual details, while simultaneously reducing the computational and memory footprint of the subsequent temporal encoder.

During training, feature-space regularization is applied by injecting zero-mean Gaussian noise into the projected embeddings. This strategy acts as a form of stochastic smoothing in the representation space, improving robustness to domain shifts and discouraging the temporal model from relying on brittle, low-level visual artifacts.

Temporal Positional Encoding

Since the transformer architecture is permutation-invariant by design, explicit positional information is required to preserve the temporal ordering of frames. We adopt a deterministic sinusoidal positional encoding scheme, following Vaswani et al. (2017), which maps each time index t to a

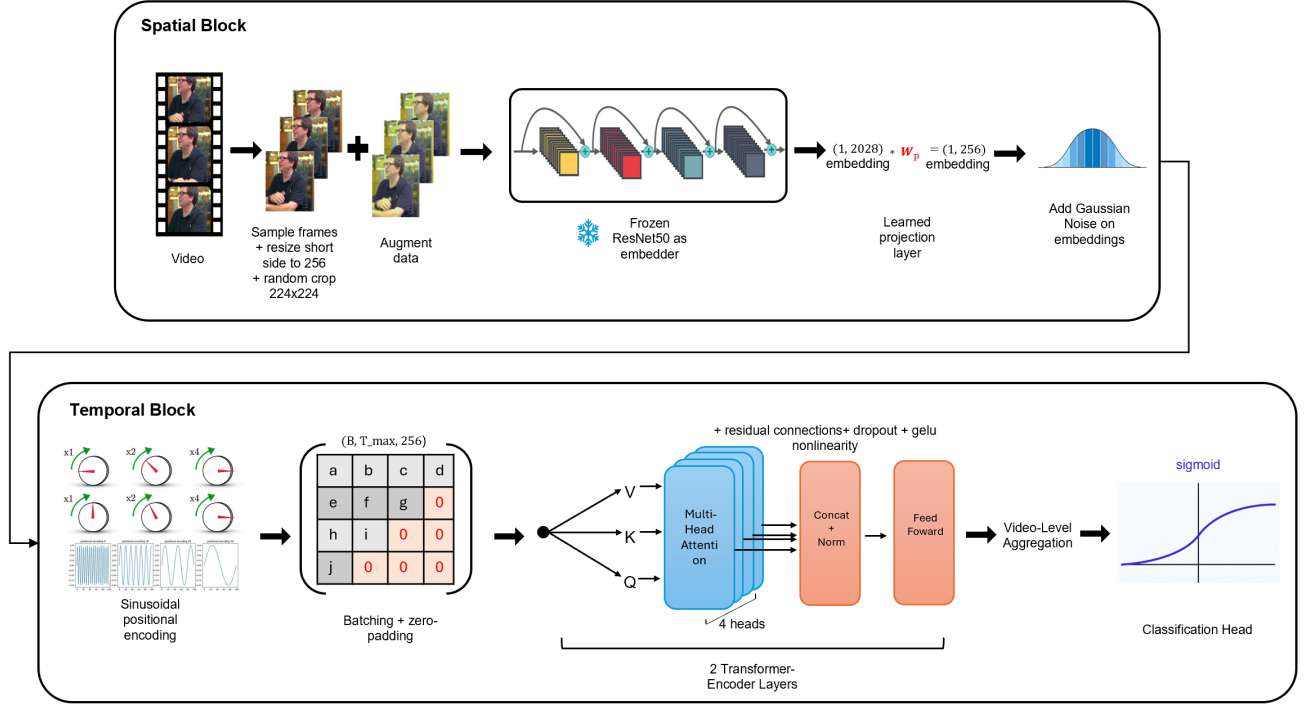


Figure 1: Full Pipeline

fixed vector $\mathbf{p}_t \in R^d$. The encoded representation is obtained via element-wise addition:

$$\tilde{\mathbf{z}}_t = \mathbf{z}_t + \mathbf{p}_t. \quad (3)$$

Each component of the positional encoding vector is defined using alternating sine and cosine functions at geometrically increasing wavelengths:

$$\mathbf{p}_t^{(2i)} = \sin\left(\frac{t}{10000^{2i/d}}\right), \quad (4)$$

$$\mathbf{p}_t^{(2i+1)} = \cos\left(\frac{t}{10000^{2i/d}}\right), \quad (5)$$

where $i \in \{0, \dots, \lfloor d/2 \rfloor - 1\}$ indexes the embedding dimensions.

This formulation allows the temporal encoder to exploit both relative and absolute frame positions when computing attention weights, enabling the modeling of long-range temporal dependencies such as escalation patterns or delayed responses in violent interactions.

Variable-Length Sequence Handling

Videos in real-world surveillance and web-based datasets exhibit substantial variation in duration, resulting in frame sequences of unequal length. To enable efficient batch processing on modern accelerators, all sequences within a batch are padded to a common temporal length using zero vectors. A corresponding binary padding mask $\mathbf{M} \in \{0, 1\}^{B \times T_{\max}}$ is maintained, where B denotes the batch size and $T_{\max}$ the maximum sequence length in the batch.

This mask is propagated through the temporal encoder to ensure that padded timesteps do not contribute to the attention computation or the final video-level representation.

Transformer-Based Temporal Encoder

Long-range temporal reasoning is performed using a multi-head self-attention transformer encoder. For a sequence of encoded frame embeddings $\tilde{\mathbf{Z}} = [\tilde{\mathbf{z}}_1, \dots, \tilde{\mathbf{z}}_T]$, each encoder block computes attention-weighted interactions across all timesteps.

For each attention head, the input is linearly projected into queries, keys, and values:

$$\mathbf{Q} = \tilde{\mathbf{Z}}\mathbf{W}_Q, \quad \mathbf{K} = \tilde{\mathbf{Z}}\mathbf{W}_K, \quad \mathbf{V} = \tilde{\mathbf{Z}}\mathbf{W}_V \quad (6)$$

Self-attention is then computed as:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathcal{M}\right)\mathbf{V} \quad (7)$$

where d_k denotes the key dimensionality and $\mathcal{M}$ represents the additive mask derived from the padding matrix, assigning large negative values to padded positions to suppress their influence.

Outputs from multiple attention heads are concatenated and passed through a position-wise feedforward network with nonlinear activation, followed by dropout, residual connections, and layer normalization. This architecture enables the model to jointly capture short-term motion cues and long-term temporal structure, such as the progression from non-violent to violent behavior across a sequence.

Video-Level Aggregation via Masked Pooling

The transformer encoder produces a sequence of context-aware frame representations $\mathbf{h}_t$. To obtain a fixed-dimensional video-level descriptor, we apply masked mean pooling:

$$\mathbf{v} = \frac{1}{\sum_{t=1}^T M_t} \sum_{t=1}^T M_t \mathbf{h}_t \quad (8)$$

where M_t denotes the validity mask for timestep t . This operation aggregates temporal information while ensuring that padded frames do not contribute to the final representation.

Classification Head and Decision Function

The pooled video representation $\mathbf{v}$ is passed through a multilayer perceptron (MLP) consisting of a linear transformation, GELU nonlinearity, and dropout regularization, followed by a final linear layer that produces a scalar logit ℓ :

$$\ell = W_c \sigma(W_h \mathbf{v} + b_h) + b_c \quad (9)$$

The predicted class probability is obtained via the sigmoid function:

$$\hat{y} = \sigma(\ell) \quad (10)$$

Training Objective and Optimization Strategy

Model parameters are optimized by minimizing the binary cross-entropy loss with logits:

$$\mathcal{L}(y, \ell) = -[y \log \sigma(\ell) + (1 - y) \log(1 - \sigma(\ell))] \quad (11)$$

This formulation provides numerical stability and directly operates on the unnormalized logit output. Gradient clipping is employed to stabilize training, particularly in the transformer components, and weight decay is applied through the AdamW optimizer to discourage overly complex solutions.

Design Rationale

The proposed architecture is motivated by a trade-off between expressive power and computational efficiency. By decoupling spatial feature extraction from temporal modeling, the framework approximates the behavior of full spatiotemporal transformers while maintaining significantly lower training and inference costs. The use of attention-based temporal encoding enables the model to capture long-range dependencies that are difficult to model using fixed-size temporal convolutions, while masking and feature-space regularization improve robustness to variable-length inputs and domain shifts commonly encountered in real-world violence detection scenarios.

This modular design further allows individual components, such as the convolutional backbone or temporal encoder, to be replaced or extended without altering the overall pipeline, facilitating adaptation to alternative datasets or deployment constraints.

Experiment

Dataset Description and Statistics

The evaluation of our hybrid architecture is conducted on the Real Life Violence Situations Dataset (Soliman et al., 2019). This dataset was specifically curated to address the lack of large-scale video data focusing on interpersonal violence in diverse real-world environments. It consists of 2,000 video clips, perfectly balanced into two categories: 1,000 "Violence" clips showing real street fights and aggressive interactions, and 1,000 "Non-Violence" clips depicting common human activities such as walking, eating, and various sports. Violence samples are primarily drawn from real-world and user-generated content, often exhibiting handheld camera motion, compression artifacts, and low-resolution characteristics. In contrast, non-violence samples include a substantial proportion of professionally produced material such as sports broadcasts and cinematic footage, resulting in comparatively stable camera motion and higher visual quality.

This systematic difference in visual and recording characteristics introduces a risk of shortcut learning, where models may exploit low-level cues such as video quality or camera stability rather than semantic indicators of violent behavior. The experimental design explicitly addresses this issue through targeted augmentation and regularization strategies described in subsequent sections.

The dataset is partitioned into disjoint training, validation, and test splits using a 70/15/15 ratio. All splits are constructed at the video level to prevent frame-level leakage between subsets.

Frame Sampling and Preprocessing

Videos are uniformly sampled at a fixed temporal rate to obtain a sequence of RGB frames for each input. Frames are spatially normalized to a consistent tensor shape to ensure compatibility with the convolutional backbone and batch processing on GPU accelerators.

Specifically, each frame is resized such that the shorter spatial dimension is scaled to 256 pixels while preserving the original aspect ratio. A spatial crop of size 224×224 is then extracted. During training, random cropping is applied to introduce spatial variability and reduce overfitting to fixed regions of the frame. For validation and testing, center cropping is employed to ensure deterministic and reproducible evaluation.

Alternative spatial normalization strategies, including reflect padding, were evaluated. While spatial cropping introduces the risk of removing localized visual cues associated with violent events, padding-based approaches preserve the full spatial extent of the frame. However, statistical analysis revealed a correlation between aspect ratio and class labels, with portrait-oriented videos occurring more frequently in the violence class and landscape-oriented videos being more prevalent in the non-violence class.

Reflect padding preserved these geometric biases and consequently increased the risk of shortcut learning, as the model could exploit aspect-ratio-related cues rather than semantically meaningful indicators of violent behavior. The adopted resize-and-crop strategy was therefore selected as a

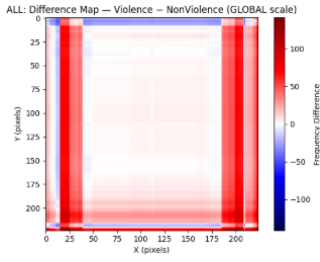


Figure 2: Difference Map of Padding Prevalence by Class

trade-off between preserving salient spatial regions and mitigating this confounding factor, while maintaining sufficient spatial coverage for robust feature extraction.

Data Augmentation and Domain Randomization

To improve robustness to domain shifts and to discourage reliance on superficial visual cues, a comprehensive data augmentation policy is applied exclusively to the training split. For each training video, both an original and an augmented variant are generated, effectively doubling the number of training samples without altering the class distribution.

Augmentation is performed at the frame level and includes the following transformations:

- JPEG compression and decompression to simulate codec artifacts and quality degradation
- Gaussian blur to model defocus and motion-induced smoothing
- Color jitter and gamma adjustment to introduce illumination variability
- Random conversion to grayscale to reduce color dependency
- Additive Gaussian noise to simulate sensor noise
- Temporal motion blur applied across consecutive frames to emulate rapid camera movement

In addition to input-space augmentation, we also experimented with feature-space regularization by injecting zero-mean Gaussian noise into the frame embeddings during training.

Batch Construction and Sequence Padding

To enable efficient parallel computation on GPU hardware, videos are processed in mini-batches rather than as individual sequences. Frame embeddings from multiple videos are stacked into a single three-dimensional tensor of shape $(B \times T_{\max} \times d)$, where B denotes the batch size, $T_{\max}$ the maximum sequence length within the batch, and d the feature dimensionality. This formulation allows attention and feedforward operations to be implemented as large, batched matrix multiplications instead of a series of independent vector operations, significantly improving computational throughput.

As described in the methodology, variable-length sequences are aligned within each batch via zero-padding.



Figure 3: Data Augmentation Channels

Training Configuration

Models are trained using the AdamW optimizer with decoupled weight decay to promote generalization. Gradient clipping is employed to stabilize optimization in the transformer layers and prevent exploding gradients in long temporal sequences.

Binary cross-entropy with logits is used as the training loss. Although the dataset is approximately class-balanced, evaluation is framed as a cost-sensitive task, reflecting the higher practical cost of missed violence events relative to false alarms in surveillance-oriented applications.

Model checkpoints are selected based on validation performance rather than training loss. Two selection criteria are explored: (i) the F1-score of the violence class, emphasizing recall and sensitivity to violent events, and (ii) balanced accuracy, which assigns equal importance to both classes and penalizes excessive false positives.

Evaluation Metrics

Performance is assessed using multiple complementary metrics to capture different operational trade-offs:

- **Precision (P):** the proportion of predicted violence samples that are true violence events
- **Recall (R):** the proportion of true violence samples correctly identified
- **F1-score:** the harmonic mean of precision and recall

$$F1 = \frac{2PR}{P + R} \quad (12)$$

- **Balanced Accuracy (BA):** the average of true positive rate and true negative rate

$$BA = \frac{1}{2} (TPR + TNR) \quad (13)$$

A fixed decision threshold of 0.5 is applied to the predicted class probability to obtain binary predictions. Confusion matrices are reported to provide a detailed breakdown of false positives and false negatives.

Main Results

On the held-out test split, the balanced model achieves consistently strong performance across all evaluation metrics, indicating a well-calibrated trade-off between sensitivity to violent events and control of false alarms. The selected checkpoint attains a balanced accuracy of 0.962, with an overall accuracy of 0.962, precision of 0.960, recall of 0.966, and an F1-score of 0.963.

Table 1: Performance metrics of the balanced model on the held-out test split.

Metric	Value
Accuracy	0.962
Balanced Accuracy	0.962
Precision	0.960
Recall	0.966
F1-score	0.963
False Positives (FP)	6
False Negatives (FN)	5

Table 2 presents the corresponding confusion matrix. The model correctly identifies 144 violent and 137 non-violent videos, with only 6 false positives and 5 false negatives. This symmetric error profile confirms that performance gains are not driven by a bias toward either class, but rather by robust discrimination across both categories.

Table 2: Confusion matrix of the balanced model on the test split.

	Predicted Non-Violence	Predicted Violence
True Non-Violence	137	6
True Violence	5	144

Cross-Dataset Generalization Analysis

To assess robustness beyond the in-distribution test set, the balanced model is evaluated on the external dataset introduced by Bianculli et al. (2020), which is designed to be particularly challenging due to the inclusion of ambiguous non-violent interactions such as friendly physical contact and crowded social scenes.

Table 3: Confusion matrix on the external dataset by Bianculli et al. (2020)

	Predicted Non-Violence	Predicted Violence
True Non-Violence	33	87
True Violence	29	201

Table 4: Classification performance on the external dataset by Bianculli et al. (2020)

Class	Precision	Recall	F1
Non-Violence	0.532	0.275	0.363
Violence	0.698	0.874	0.776
Accuracy	0.669		

Although the model maintains high sensitivity to violent events, performance on non-violent samples degrades substantially. The external dataset contains visually and semantically different non-violent scenarios, including sports footage and professionally produced media with stable camera motion, letterboxing, and distinctive visual styles. This suggests that the dominant performance bottleneck lies in the distribution of the training data rather than in the temporal modeling architecture itself. The largest leverage for improvement is therefore expected to come from expanding and diversifying the non-violent training corpus to better reflect socially ambiguous real-world interactions.

Implementation Details

All experiments are conducted using T4-GPU-accelerated training. Model training, evaluation, and logging are implemented in a modular pipeline that supports reproducibility through fixed random seeds and deterministic data splits. Training logs, confusion matrices, and final metrics are recorded for each run to facilitate comparative analysis across model variants and hyperparameter configurations.

Conclusion

This work presents a modular video-based violence detection framework that decouples spatial feature extraction from long-range temporal reasoning through a frozen ResNet backbone and a transformer-based encoder. The proposed design achieves strong in-distribution performance under balanced evaluation criteria while maintaining computational efficiency and architectural flexibility. Cross-dataset evaluation on a challenging external benchmark highlights a pronounced sensitivity to domain shift, particularly in visually and semantically ambiguous non-violent scenarios, suggesting that generalization is primarily constrained by training data diversity rather than limitations in the temporal modeling architecture. These findings motivate future work on large-scale, socially diverse pretraining, domain-adaptive learning strategies, and multimodal extensions to improve robustness in unconstrained real-world environments.

References

- [2021] Arnab, A.; Dehghani, M.; Heigold, G.; Sun, C.; Lučić, M.; and Schmid, C. 2021. Vivit: A video vision transformer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 6836–6846.
- [2021] Bertasius, G.; Wang, H.; and Torresani, L. 2021. Is space-time attention all you need for video understanding? In *Proceedings of the International Conference on Machine Learning (ICML)*, 813–824.
- [2020] Bianculli, M.; Falcionelli, N.; Sernani, P.; Tomassini, S.; Contardo, P.; Lombardi, M.; and Dragoni, A. F. 2020. A dataset for automatic violence detection in videos. *Data in Brief* 33: 106587.
- [2017] Carreira, J., and Zisserman, A. 2017. Quo vadis, action recognition? A new model and the kinetics dataset. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 6299–6308.
- [2019] Girdhar, R.; Carreira, J.; Doersch, C.; and Zisserman, A. 2019. Video action transformer network. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 244–253.
- [2012] Hassner, T.; Itcher, Y.; and Kliper-Gross, O. 2012. Violent flows: Real-time detection of violent crowds. In *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 1–6.
- [2016] He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.
- [2015] Tran, D.; Bourdev, L.; Fergus, R.; Torresani, L.; and Paluri, M. 2015. Learning spatiotemporal features with 3d convolutional networks. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 4489–4497.
- [2017] Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, Ł.; and Polosukhin, I. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 5998–6008.